Pronob Kumar Barman

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PROFESSIONAL EXPERIENCE

University of Maryland, Baltimore County
Teaching Assistant, Information Systems 01/2022 - Present

- Guided graduate students in applying **machine learning techniques** (e.g., regression models, data models, decision trees, randome forest, XGBoost, SVMs, neural networks) to real-world projects, emphasizing feature engineering, hyper-parameter tuning, and model evaluation.
- Led advanced sessions on **deep learning architectures**, optimization, and causal inference.

Technuf LLC
Research Scientist, Intern 06/2024 - 08/2024

- Developed advanced Generative AI models, integrating **LLMs (GPT, BERT)** with SIEM systems to improve cybersecurity measures, reducing security breaches by identifying threats faster and more accurately.
- Achieved a **25% improvement in threat detection accuracy** through model optimization.

Data Scientist, Intern 08/2022 - 12/2022

- Led a data-driven **Medicaid analysis project using ML models**, reducing operational costs by 55% and increasing revenue by 80%.
- Integrated outcomes into business strategies, strengthening Technuf LLC's healthcare analytics services.

University of Kentucky
Teaching Assistant, Department of Statistics 08/2019 - 05/2021

- Assisted undergraduate students in understanding and applying **key statistical models**, including linear regression, logistic regression, ANOVA, and time series analysis. Guided students through real-world data analysis projects with a focus on data modeling, anomaly detection, forecasting, and data management across qualitative and quantitative datasets.
- Provided instruction on **probability theory** and statistical inference.

Bangladesh Bank
Deputy Director, Chief Economist's Unit 04/2018 - 12/2021

- Directed economic policy research and statistical projects, improving **national economic models by 65%**, which informed key financial decisions and enhanced financial stability.

EDUCATION

University of Maryland, Baltimore County
PhD in Information Systems, CGPA: 3.85/4.00 01/2022 - Present
Research Focus: Generative Models, NLP, Recommendation Systems, UX

University of Kentucky
MS in Statistics, CGPA: 3.85/4.00 08/2019 - 05/2021
Advanced Theoretical Focus: Probability Theory, Statistical Inference, Experimental Design

University of Dhaka
MS in Statistics, CGPA: 3.46/4.00 10/2015 - 01/2017
BS in Statistics, CGPA: 3.46/4.00 01/2011 - 08/2015

SKILLS

- **Programming Languages:** Python, R, SQL, SAS, Java, JavaScript, Matlab
- **Data Science Libraries:** NumPy, SciPy, Pandas, NLTK, SymPy, IPython, PyTables, StatsModels, NetworkX, Mayavi
- **ML Tools:** TensorFlow, PyTorch, Keras, Scikit-learn
- **Generative AI:** LangChain, OpenAI API, Hugging Face, Pinecone, ChromaDB, MongoDB
- **Code Debugging:** Profiling tools (e.g., cProfile, Py-Spy)
- **Data Manipulation:** Hadoop, Spark, AWS Cloud, Oracle SQL, PostgreSQL
- **Data Visualization:** Matplotlib, Seaborn, Plotly, ggplot2, Tableau, Power BI
- **Tools:** ConnectWise SIEM & SOC, Git, SPSS, LaTeX, Visual Studio IDE, Microsoft Project, Teams, Outlook, Qualtrics
- **Soft Skills:** Leadership, Effective Communication, Problem-solving

PHD RESEARCH PROJECT

Online Healthcare Support Group Formation and Recommendation System

- Developed an innovative automated support group formation system by leveraging two proposed **generative models**, Group-specific Dirichlet Multinomial Regression (gDMR) and Group-specific Structural Topic Model (gSTM), to optimize user-group assignments based on personalized content, demographic features, and interaction data, resulting in enhanced group coherence, stability, and user engagement.

- Engineered advanced **recommendation systems** in Python, Java, and R, incorporating **topic modeling** and node embeddings from user interaction data, improving accuracy in personalized support group suggestions by 75%.
- Led a comprehensive **user experience survey** to evaluate the perceived benefits, essential features, and privacy concerns of the system, shaping future iterations of the support platform based on user feedback regarding group size and communication preferences.
- Achieved notable improvements in topic coherence, log-likelihood, and fairness metrics, setting new benchmarks for personalized healthcare support systems through real-world data application.
- Released an R package for gSTM and shared the Python implementation of gDMR, enabling other researchers and developers to implement these models in their own projects, enhancing scalability and adoption across various domains.

OTHER PROJECTS

LLM Generative AI Project with LangChain and OpenAI API Github Link

- Built a comprehensive Generative AI system using LangChain and OpenAI API, integrating Pinecone and ChromaDB for advanced semantic search and retrieval-augmented generation.

End-to-End Medical Chatbot with Llama 2 and Hugging Face API Github Link

- Developed a real-time medical chatbot utilizing Meta's Llama 2 and Hugging Face API, enhancing patient interactions with LangChain integration for improved conversational flow and accuracy.

Predictive Policing with GAN

- Implemented a Generative Adversarial Network (GAN) model to identify and mitigate bias in predictive policing algorithms, achieving a 20% reduction in false-positive rates.

Classification of Medical Specialties from Transcriptions

- Achieved 98% classification accuracy by developing a SMOTE-enhanced machine learning model for classifying medical specialties from transcription data.

Cognitive and Affective State Recognition in iVR Using Multimodal Signals Github Link

- Developed inverse inference models mapping neuro-physiological signals to cognitive and affective states in immersive virtual reality (iVR), analyzing data from 10 participants to assess the impact of design decisions on model performance.

PUBLICATIONS

- Walid, M. A. A., Devnath, M. K., Muiz, B., Melon, M. M. H., **Barman, P. K.**, & Barman, P. K. (2024). "Efficient Graph-Based Intrusion Detection for CAN Bus Security with Minimal Training Data." *Proceedings of the 6th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI 2024)*, 2024.
- Rachael, K., Reynolds, T. L., Yus R. & **Barman, P. K.** (2025). "Acceptance of Occupancy Dashboards among Undergraduate Students: Towards Everyday Decision-making Infrastructure that can be Leveraged in Pandemics." *CHI Conference on Human Factors in Computing Systems*, 2025. (Under Review)
- Balogun, Z. A., **Barman, P. K.**, & Reynolds, T. L. (2025). "Cultivating a Personalized Digital Health Ecosystem: Insights from a Mixed Methods Survey of U.S. Adults." *CHI Conference on Human Factors in Computing Systems*, 2025. (Under Review)
- Balogun, Z. A., **Barman, P. K.**, Onwumbiko, B. K., & Reynolds, T. L. (2024). "User-generated Data for Different Types of Mobile-Personal Health Records: A Computationally-guided Qualitative Analysis Approach." *AMIA Annual Symposium, 2024*.
- Faruq, M. O., Walid, M. A. A., Baowaly, M. K., Devnath, M. K., Ejaz, M. S., **Barman, P. K.**, & Sattar, A. S. (2024). "Shielding Privacy: A Technique of Extenuating Composition Attacks in Various Independent Data Publication." *Bulletin of Electrical Engineering and Informatics, 13(4), 2677-2686*.
- **Barman, P. K.** (2023). "Detecting Abnormal Health Conditions in Smart Home Using a Drone." *arXiv preprint arXiv:2310.05012*.

PEER REVIEWER

- NeurIPS 2024 Workshop on Behavioral ML
- AMIA 2024 Annual Symposium